

Platform per operation:

Operation	Representative Platform	Why this is the right choice
Chat with AI	ChatGPT	Direct, widely used example of LLM interaction; clearly triggers remote inference for each prompt
Real-time translation (10–30 sec)	Google Translate	Industry standard for real-time voice + camera translation; clearly cloud-dependent and multi-modal
Summarize PDF document (10 pages)	Claude	Strong association with long-context document analysis; represents heavy inference workloads and large input processing
Remove background from image	Adobe Express	Clear, single-purpose computer vision tool; easy to understand and strongly tied to cloud processing

Data centers:

Prompt(Chat with AI):

Consumer ChatGPT + API Customers

These are the large, established, high-capacity hubs. Consumer ChatGPT at global scale requires massive GPU pools, and these regions have the infrastructure density to support it.

Region	City	Country	Why both
East US	Northern Virginia	USA	Oldest and largest OpenAI inference hub globally
East US 2	Virginia	USA	Primary home of OpenAI's production workloads on Azure

South Central US	Texas	USA	Major US capacity node, Stargate infrastructure nearby
West US	California	USA	Close to OpenAI HQ, early inference region
Sweden Central	Stockholm	Sweden	Primary EU hub, largest EU inference capacity
France Central	Paris	France	Major EU hub, confirmed Data Zone anchor
West Europe	Amsterdam	Netherlands	Oldest EU Azure OpenAI node, high capacity
UK South	London	UK	Large hub, confirmed for consumer EU routing
Japan East	Tokyo	Japan	Primary APAC hub, massive user base in region
Australia East	Sydney	Australia	Primary Oceania hub

Primarily API Customers

These regions exist mainly to give API developers geographic flexibility — for compliance, latency, or data residency reasons. They have real GPT inference capacity but are unlikely to carry significant consumer ChatGPT load.

Region	City	Country	Why primarily API
North Central US	Illinois	USA	Secondary US node, API overflow capacity
West US 3	Arizona	USA	Newer region, limited consumer routing evidence
Germany West Central	Frankfurt	Germany	Compliance-driven deployments, GDPR enterprise focus
Switzerland North	Zürich	Switzerland	Sovereign compliance use cases, small capacity
Norway East	Oslo	Norway	Added for Nordic enterprise data residency needs

Poland Central	Warsaw	Poland	Recent addition, primarily Central-Eastern EU enterprise
Korea Central	Seoul	South Korea	Regional API coverage for Korean enterprise market
South Africa North	Johannesburg	South Africa	African market API access, small consumer base
Brazil South	São Paulo	Brazil	LATAM API coverage, consumer routing unconfirmed

BARCELONA:

Prompt(Chat with AI):

ChatGPT runs on **Microsoft Azure** (OpenAI's primary infrastructure partner). Global Standard deployments use the global infrastructure of Azure, dynamically routing customer traffic to the data center with the best availability for each inference request. This means the actual destination is never fixed — it depends on load, latency, and capacity at the moment of the request.

Data centers:

Azure Region	Physical City	Distance from BCN	Notes	Realistic
West Europe	Amsterdam, NL	~1,400 km	Oldest Azure OpenAI EU hub; historically primary for European inference	Legacy problem
France Central	Paris, FR	~1,000 km	Confirmed OpenAI support; geographically closest major hub	Primary

German y West Central	Frankfurt, DE	~1,500 km	Growing OpenAI capacity	Secondary
Sweden Central	Stockholm, SE	~2,500 km	The primary EU Data Zone region for newer GPT models Microsoft Learn ; higher latency from Barcelona	Primary
UK South	London, UK	~1,400 km	Confirmed OpenAI support, but post-Brexit GDPR routing is less common	Secondary
Switzerl and North	Zürich, CH	~1,300 km	Sovereign compliance use cases	Secondary

This table maps the Azure infrastructure regions that could realistically receive and process a ChatGPT request originating from Barcelona. Because OpenAI runs on Microsoft Azure and routes traffic dynamically based on load and latency, no single destination is fixed — instead, a pool of candidate regions exists. Each row identifies the Azure region by its official name, its physical city location, its approximate network distance from Barcelona, and a note explaining its role in OpenAI's European inference infrastructure. The final column assigns a routing likelihood: Primary for regions with full model support and strong capacity that would typically be selected first, Secondary for confirmed but less frequently used destinations, and Legacy for regions that were historically dominant but have since been deprioritised as newer infrastructure has come online. Together the rows establish that Paris and Stockholm are the most probable current destinations, while Amsterdam — despite being the oldest European Azure OpenAI hub — now represents a legacy case.

<https://azure.microsoft.com/en-us/explore/global-infrastructure/>

<https://learn.microsoft.com/en-us/azure/foundry/foundry-models/concepts/models-sold-directly-by-azure?tabs=global-standard-aoai%2Cglobal-standard&pivots=azure-openai>

Energy consumption for each data center:

Destination	Total Electricity	CO ₂	Water
Paris	0.00036–0.0024 kWh	0.008–0.053 g	0.11–0.72 mL
Stockholm	0.00035–0.0023 kWh	0.008–0.053 g	0.06–0.41 mL
Frankfurt	0.00036–0.0024 kWh	0.126–0.840 g	0.11–0.72 mL
Zürich	0.00035–0.0023 kWh	0.015–0.097 g	0.09–0.58 mL
Amsterdam	0.00036–0.0024 kWh	0.094–0.629 g	0.11–0.72 mL
London	0.00036–0.0024 kWh	0.045–0.298 g	0.09–0.60 mL

This table shows the full environmental cost of executing a single ChatGPT prompt from Barcelona, calculated separately for each of the six candidate data center destinations. The same core compute energy range of 0.0003 to 0.002 kWh applies in all cases, since the model performs identical operations regardless of location. What changes across destinations is the total electricity after applying each facility's PUE multiplier, and more dramatically, the CO₂ and water figures, which are determined by the carbon intensity of the local electricity grid and the cooling water requirements of the local climate. Paris and Stockholm produce the lowest CO₂ per action — both under 0.053 g at the upper end — because France's nuclear-heavy grid and Sweden's hydro and wind mix are among the cleanest in Europe. Frankfurt produces by far the highest CO₂, reaching up to 0.840 g, because Germany's grid still carries a significant fossil fuel share. Amsterdam and London sit in the middle range, with London notably cleaner than Amsterdam thanks to the UK's rapid wind energy expansion. Zürich is the second cleanest after Paris and Stockholm due to Switzerland's hydro and nuclear generation mix, though it sits outside EU regulatory routing preferences. The water column follows climate logic: Stockholm uses the least, benefiting from cold northern temperatures that allow air-based cooling, while warmer or more humid locations require more water-intensive cooling systems.

Translation:

Cloud Translation Basic API and the consumer-facing Google Translate use global endpoints — data cannot be configured to remain in a specific region for processing. This means unlike ChatGPT with its EU Data Zone, standard Google Translate can technically route anywhere in the world, including US data centers. The only constraint is latency — which for real-time streaming translation is the dominant factor.

Correction — Consumer Google Translate vs Cloud Translation API. Google Translate the consumer app runs on Google's own internal backbone and TPU infrastructure — it routes to Google-owned campuses, not necessarily to the public GCP colocation regions. Only the Cloud Translation API for developers uses those GCP regions.

Data centers:

Location	Distance from BCN	Realistic?	Why
Belgium (St. Ghislain)	~1,100 km	Primary	Oldest, largest EU Google campus, core routing hub
Netherlands (Eemshaven)	~1,500 km	Primary	Largest known Google EU facility
Germany (Hanau/Frankfurt)	~1,500 km	Secondary	Smaller owned campus, confirmed EU presence
Finland (Hamina)	~3,000 km	Unlikely	Major campus but too far — latency breaks real-time streaming

This table maps the infrastructure locations that could realistically receive and process a real-time Google Translate request originating from Barcelona. Unlike ChatGPT, which runs on Microsoft Azure, Google Translate is a first-party Google product that runs on Google's own internal backbone network and TPU infrastructure. The relevant destinations are therefore Google-owned hyperscale campuses, not the public Google Cloud Platform regions, which are designed for third-party developer use. Each row identifies the physical facility, its ownership type, its approximate distance from Barcelona, and a routing likelihood. The two primary destinations are Belgium and the Netherlands, which host Google's oldest, largest, and most established European campuses. Germany represents a secondary owned option. Finland, while a major Google-owned campus, is classified as unlikely due to its distance of approximately 3,000 km from Barcelona — a latency that would compromise the real-time streaming experience that translation requires. The Paris entry, though geographically closest, is marked as a leased colocation facility operated by Global Switch rather than a Google hyperscale campus, making it a possible but secondary routing destination. The original PDF assignment of Barcelona to Madrid has been revised: Google has no owned data center in Spain, and the Madrid Google Cloud region relies on leased third-party infrastructure with limited AI inference capacity.

Energy consumption for each data center:

Destination	Total Electricity	CO ₂	Water
Belgium	0.00109–0.0109 kWh	0.148–1.48 g	0.24–2.40 mL
Netherlands	0.00109–0.0109 kWh	0.282–2.82 g	0.22–2.18 mL
Germany	0.0011–0.011 kWh	0.408–4.08 g	0.33–3.30 mL
Finland	0.00109–0.0109 kWh	0.093–0.93 g	0.16–1.64 mL

This table shows the full environmental cost of executing a single real-time translation session of 10 to 30 seconds from Barcelona, calculated separately for each candidate destination. The core compute energy range of 0.001 to 0.01 kWh is notably higher than for a chat prompt, reflecting the continuous multi-model pipeline — speech recognition, vision processing, and language inference running simultaneously throughout the session. A critical distinction from the ChatGPT calculations is the PUE applied: Google's owned campuses in Belgium, the Netherlands, and Finland all achieve a verified PUE of 1.09, compared to the 1.2 used for Azure facilities, meaning Google's infrastructure adds only 9% overhead energy per unit of computation rather than 20%. This difference is meaningful at scale. Belgium produces moderate CO₂ between 0.148 and 1.48 grams per session, with very low water consumption of 0.24 to 2.40 millilitres thanks to canal-based cooling. The Netherlands produces slightly higher CO₂ at 0.282 to 2.82 grams due to its gas-heavier grid, but uses even less water through seawater cooling. Germany is the worst CO₂ outcome at up to 4.08 grams per session, driven by its fossil-heavy electricity mix. Finland would produce the lowest CO₂ of the realistic owned destinations at 0.093 to 0.93 grams, but its distance makes it unsuitable. Paris colocation produces the lowest CO₂ figures of all — 0.026 to 0.264 grams — because France's nuclear grid is exceptionally clean, but this must be read alongside the fact that the facility is not Google's own infrastructure and therefore carries a higher PUE of 1.20. The water column reflects each location's cooling method directly: seawater and canal cooling in northern Europe use significantly less water than conventional mechanical cooling systems further south.

Summarize pdf:

Data centers:

City	Country	Distance from BCN	Why	Realistic
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Dublin	Ireland	~1,800 km	Oldest and largest AWS EU region; highest inference capacity in Europe; most established hub for Anthropic's European traffic	Primary
Paris	France	~1,000 km	Major AWS EU region confirmed for Claude inference; geographically close; original PDF assignment	Primary
Frankfurt	Germany	~1,500 km	Major AWS EU hub with confirmed Claude support; high capacity and established routing	Primary
Zaragoza	Spain	~300 km	Closest AWS region to Barcelona; confirmed in Bedrock EU inference profile; but Anthropic's own claude.ai routing to this region is not publicly documented	Possible
Milan	Italy	~700 km	Confirmed in Bedrock EU inference profile; relatively close; but same caveat — not confirmed specifically for claude.ai consumer routing	Possible
Stockholm	Sweden	~2,500 km	Confirmed in Bedrock EU inference profile; but distance makes it a lower priority and claude.ai routing not confirmed	Unlikely

This table maps the AWS infrastructure locations that could realistically receive and process a Claude PDF summarisation request originating from Barcelona. Unlike ChatGPT, which runs on Microsoft Azure, and Google Translate, which runs on Google's own hyperscale campuses, Claude is deployed on Amazon Web Services — Anthropic's primary cloud infrastructure partner. For European users, requests are routed within the EU inference geography, which defines a pool of six AWS regions across which traffic is distributed dynamically based on available capacity at the time of the request. Each row identifies the physical city, the country, the approximate distance from Barcelona, the reasoning behind its inclusion, and an assessment of how likely it is to receive the request. Dublin, Paris, and Frankfurt are classified as primary destinations because they are the oldest, largest, and most established AWS EU regions with confirmed Claude support. Zaragoza is the geographically closest AWS region to Barcelona at approximately 300 km and is confirmed in the Bedrock EU inference profile, but its role in Anthropic's own consumer-facing routing for claude.ai is not publicly documented, making it a possible but unconfirmed destination. Milan and Stockholm carry the same caveat. US East Virginia is listed as an edge case only — technically reachable via the global endpoint when EU capacity is saturated, but strongly deprioritised for European users due to GDPR data residency requirements. It is worth noting that PDF summarisation, unlike real-time translation, is not latency-sensitive: the model requires several seconds to process a long document

regardless of routing distance, which means AWS has more flexibility to assign the request based on capacity rather than proximity alone.

Energy consumption for each data center:

Destination	Total Electricity	CO ₂	Water
Dublin	0.00224–0.0224 kWh	0.573–5.73 g	0.78–7.84 mL
Paris	0.00224–0.0224 kWh	0.049–0.493 g	0.67–6.72 mL
Frankfurt	0.00224–0.0224 kWh	0.831–8.31 g	0.78–7.84 mL
Zaragoza	0.00224–0.0224 kWh	0.280–2.80 g	0.90–8.96 mL
Milan	0.00224–0.0224 kWh	0.697–6.97 g	0.78–7.84 mL
Stockholm	0.00224–0.0224 kWh	0.052–0.515 g	0.45–4.48 mL

This table shows the full environmental cost of executing a single Claude PDF summarisation request from Barcelona, calculated separately for each of the seven candidate destinations. The core compute energy range of 0.002 to 0.02 kWh is the highest of the four AI operations in this project, reflecting the memory-intensive nature of processing a large document through a long-context language model. The total electricity range of 0.00224 to 0.0224 kWh is consistent across all destinations because AWS European data centers share a verified average PUE of 1.12 — slightly less efficient than Google's owned campuses at 1.09, but better than the 1.20 applied to Azure colocation facilities. What varies dramatically across destinations is the CO₂ output, which is determined entirely by the carbon intensity of each country's electricity grid. Paris produces the least CO₂ at 0.049 to 0.493 grams per request, because France's nuclear-heavy grid has a carbon intensity of just 22 gCO₂/kWh — the cleanest grid of any destination in this entire project. Frankfurt is the worst-performing EU destination at 0.831 to 8.31 grams, driven by Germany's continued reliance on coal and gas. Dublin sits surprisingly

high at 0.573 to 5.73 grams despite being the primary AWS EU hub, because Ireland's grid still depends heavily on natural gas and has not yet built out sufficient renewable capacity to offset the rapid growth in data center demand. Zaragoza, if used, would perform moderately well at 0.280 to 2.80 grams thanks to Spain's increasingly renewable grid. Stockholm would be the second cleanest destination after Paris at 0.052 to 0.515 grams, but its routing probability is lower. US East Virginia produces the highest CO₂ of any destination at 0.891 to 8.91 grams, reinforcing the environmental as well as legal argument for keeping European AI requests within EU infrastructure. Water consumption follows climate logic: Zaragoza and US East have the highest water figures due to warmer operating environments, while Stockholm uses the least.

Background removal:

Data centers:

City	Country	Distance from BCN	Why	Realistic
Ashburn, Virginia	USA	~6,800 km	AWS us-east-1 — confirmed primary Firefly AI inference region per Adobe's own security documentation	Primary — confirmed
The Dalles / Portland, Oregon	USA	~9,400 km	AWS us-west-2 — confirmed secondary Firefly inference region in Adobe's documentation	Secondary — confirmed
Dublin, Ireland	Ireland	~1,800 km	AWS eu-west-1 — confirmed Adobe EMEA edge server location; handles caching and routing but not the AI inference itself	Edge/cache only
Paris, France	France	~1,000 km	AWS eu-west-3 — confirmed Adobe EMEA edge server location; same role as Dublin	Edge/cache only

This table maps the infrastructure locations that process a background removal request made through Adobe Express by a user in Barcelona. Adobe Express is a product of Adobe, headquartered in San Jose, California, and runs on Amazon Web Services as its primary cloud infrastructure. The background removal feature is powered by Adobe Firefly and Adobe Sensei, which are Adobe's proprietary AI engines. A critical distinction emerges when examining how Adobe's infrastructure is actually structured for European users: there are two separate layers operating in parallel. The first layer consists of edge servers in Ireland and France, which handle initial request routing, image caching, and response delivery — these are the closest points of contact for a European user. The second layer, where the actual AI computation takes place, is

located in the United States. Adobe's own security documentation for Firefly Services explicitly confirms that AI inference — including the background removal operation — is processed in AWS data centers in the US-East and US-West regions, corresponding to Northern Virginia and Oregon respectively. This means that while Ireland and Paris appear to be European touchpoints, they perform no background removal computation whatsoever. The image travels from Barcelona to an Irish or French edge server, is then forwarded across the Atlantic to Virginia or Oregon for processing, and the result is returned via the same European edge layer. Virginia is the primary inference destination, with Oregon serving as a secondary region. This transatlantic routing pattern makes Adobe Express the second operation in this project, alongside PhotoRoom, where a European user's AI computation is confirmed to occur in the United States rather than in Europe.

Energy consumption for each data center:

Destination	Total Electricity	CO ₂	Water
Virginia	0.00057–0.0057 kWh	0.198–1.98 g	0.23–2.28 mL
Oregon	0.00057–0.0057 kWh	0.114–1.14 g	0.17–1.71 mL
Dublin	edge/cache only	negligible	negligible
Paris	edge/cache only	negligible	negligible

This table shows the full environmental cost of executing a single background removal request through Adobe Express from Barcelona, calculated for the two active inference destinations. The core compute energy range of 0.0005 to 0.005 kWh is the lowest of all four AI operations in this project, reflecting the relatively efficient nature of single-pass computer vision segmentation — a focused model that isolates the subject from the background in one operation without the sustained multi-model pipeline of translation or the long-context processing of document summarisation. The total electricity range of 0.00057 to 0.0057 kWh is consistent across both US destinations because AWS American data centers share a verified average PUE of 1.14, slightly higher than Google's owned campuses but comparable to the other AWS-hosted operations in this project. The key difference between Virginia and Oregon lies entirely in their electricity grids. Virginia produces between 0.198 and 1.98 grams of CO₂ per image, reflecting a grid that still draws heavily on natural gas with only 28% coming from low-carbon sources including nuclear. Oregon performs significantly better at 0.114 to 1.14 grams, benefiting from a grid where 41% of generation comes from hydropower and renewables collectively provide 62% of total output. Water consumption follows climate logic: Virginia's hot and humid East Coast environment requires more intensive cooling at 0.23 to 2.28 millilitres per image, while Oregon's cooler Pacific Northwest climate reduces this to 0.17 to 1.71 millilitres. Ireland and Dublin are

listed but carry no energy figures for this operation because they perform no AI inference — they act purely as edge routing and caching nodes, contributing only negligible overhead that is not attributable to the background removal computation itself. Comparing these figures to the other three operations in the project, background removal is the lightest in terms of electricity consumed per action — but its transatlantic routing means the CO₂ cost is higher than it would be if the same computation were processed in a European data center, where grids such as France's at 22 gCO₂/kWh or Sweden's at 23 gCO₂/kWh would produce only a fraction of the emissions generated in Virginia.

RIYADH:

Prompt:

Data centers:

Background removal:

Data centers:

City	Countr y	Distance from Riyadh	Why	Realistic?
Ashburn, Virginia	USA	~11,000 km	AWS us-east-1 — current confirmed primary Firefly inference region; same as Barcelona but 4,200 km further	Current — confirmed
The Dalles, Oregon	USA	~13,500 km	AWS us-west-2 — confirmed secondary Firefly region	Current secondary — confirmed
Abu Dhabi / Dubai	UAE	~1,400 km	AWS me-central-1 — closest operational AWS region; likely edge/cache role only; no confirmed Firefly inference	Edge/cache only

Bahrain	Bahrain	~1,200 km	AWS me-south-1 — established Middle East AWS hub; possible edge routing	Edge/cache only
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Energy consumption for each data center:

Destination	Scenario	Total Electricity	CO ₂	Water
Virginia	Current	0.00057–0.0057 kWh	0.198–1.98 g	0.23–2.28 mL
Oregon	Current	0.00057–0.0057 kWh	0.114–1.14 g	0.17–1.71 mL

SINGAPORE:

“In Singapore, most AI operations are processed locally due to the high concentration of hyperscale data centers, minimizing geographic data transfer while concentrating energy demand within the city.”

Background removal:

Data centers:

City	Country	Distance from Singapore	Why	Realistic
The Dalles / Portland, Oregon	USA	~11,600 km	AWS us-west-2 — most likely primary Firefly inference for Singapore users given Pacific geography	Primary
Ashburn, Virginia	USA	~15,300 km	AWS us-east-1 — confirmed secondary Firefly inference region	Secondary

Singapore	Singapore	0 km	AWS ap-southeast-1 — confirmed Adobe edge server location; handles routing and caching only; no Firefly inference	Edge/cache only
Tokyo	Japan	~5,300 km	AWS ap-northeast-1 — confirmed Adobe storage location; possible future inference expansion but not currently confirmed	Storage only

This table maps the infrastructure locations that process a background removal request made through Adobe Express by a user in Singapore. The routing architecture for Singapore follows the same confirmed pattern as Barcelona and Riyadh: Adobe Firefly Services, which powers the background removal operation, processes all inference exclusively in AWS data centers in the US-East and US-West regions, as documented in Adobe's own security fact sheet. Singapore itself is confirmed as an Adobe APAC edge server location, meaning it plays a real role in the request journey — receiving the image from the user, handling initial routing, and caching the response on return — but it performs no AI computation whatsoever. Tokyo is similarly listed as an Adobe cloud storage location for APAC users, handling data persistence but not inference. The critical geographic distinction for Singapore compared to the other three cities in this project is that its Pacific position makes Oregon (us-west-2) the closer and therefore more likely primary inference destination, at approximately 11,600 km away, versus Virginia (us-east-1) which sits approximately 15,300 km away. This makes Singapore's confirmed minimum routing distance the longest of any city in this project, including Riyadh's current 11,000 km journey. This is particularly striking given that Singapore is one of the most data-centre-dense cities on the planet, hosting one of AWS's largest Asia-Pacific regions and enormous local cloud infrastructure across multiple providers. Despite all of this local capacity, Adobe's inference for a Singapore user travels further than for any other city studied in this project, illustrating perhaps more forcefully than any other example that the density of infrastructure surrounding a user tells you nothing about where their AI computation is actually performed.

Energy consumption for each data center:

Destination	Scenario	Total Electricity	CO ₂	Water
Oregon	Primary	0.00057–0.0057 kWh	0.114–1.14 g	0.17–1.71 mL
Virginia	Secondary	0.00057–0.0057 kWh	0.198–1.98 g	0.23–2.28 mL

This table shows the full environmental cost of executing a single background removal request through Adobe Express from Singapore, calculated for the two active inference destinations. The core compute energy range of 0.0005 to 0.005 kWh and the total electricity range of 0.00057 to 0.0057 kWh are identical to Barcelona and Riyadh, because all three cities route to the same AWS infrastructure running the same Firefly models on the same hardware with the same PUE of 1.14. The image does not know where it came from. What changes for Singapore — compared to Barcelona and Riyadh — is the primary inference destination. Whereas Barcelona and Riyadh route primarily to Virginia, Singapore's Pacific geography makes Oregon the more likely first-choice destination. This geographic shift turns out to be environmentally advantageous entirely by accident: Oregon's grid, dominated by hydroelectric power at 41% and renewables at 62% of total generation, produces only 0.114 to 1.14 grams of CO₂ per image at the primary destination — the lowest current figure of any city in this project. Virginia, the secondary destination, produces 0.198 to 1.98 grams. Singapore and Tokyo are listed but carry no energy figures because they perform no inference computation — they act purely as edge routing and storage nodes whose contribution to the background removal operation is negligible. Placing this result alongside the other three cities reveals the central finding of the Adobe Express operation across this project: three cities on three different continents — Barcelona at 6,800 km, Riyadh at 11,000 km, and Singapore at 11,600 km — all produce essentially comparable CO₂ footprints for the exact same operation, because they all end up in the same data centers running on the same grids. The only meaningful environmental variable is not the user's location, not the distance the data travels, and not the local infrastructure density. It is solely the carbon intensity of the electricity grid powering the data center where the computation lands — a data center the user never chose, cannot see, and in most cases does not know exists.

NAIROBI:

“In Nairobi, most AI operations are processed in external regional hubs due to limited local hyperscale infrastructure, resulting in longer data routes and dependency on external compute regions.”

Background removal:

Data centers:

City	Countr y	Distance from Nairobi	Why	Realistic
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Ashburn, Virginia	USA	~11,900 km	AWS us-east-1 — confirmed primary Firefly inference; Atlantic-facing geography from East Africa makes this the logical primary	Primary — confirmed
The Dalles, Oregon	USA	~14,600 km	AWS us-west-2 — confirmed secondary Firefly inference region	Secondary — confirmed
Dublin, Ireland	Ireland	~7,300 km	Nearest confirmed Adobe EMEA edge server ; Nairobi may route through here before forwarding to Virginia	Possible intermediate edge

Energy consumption for each data center:

Destination	Scenario	Total Electricity	CO ₂	Water
Virginia	Primary	0.00057–0.0057 kWh	0.198–1.98 g	0.23–2.28 mL
Oregon	Secondary	0.00057–0.0057 kWh	0.114–1.14 g	0.17–1.71 mL

MAIN COMPARISONS:

Background removal:

City	Actual distance to inference	CO ₂ per image
Barcelona	6,516 km → Virginia	0.198–1.98 g
Riyadh (current)	10,862 km → Virginia	0.198–1.98 g
Nairobi	12,176 km → Virginia	0.198–1.98 g

Singapore	13,181 km → Oregon	0.114–1.14 g
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